

NFR TESTING

Version: 2.0

Last Updated: 03 September 2026



Prepared by



TESTING THE '/'

10-25 VUs → baseline load

```
stages: [  
  { duration: '30s', target: 10},  
  { duration: '1m', target: 10},  
  { duration: '30s', target: 25},  
  { duration: '1m', target: 25},  
  { duration: '30s', target: 0},  
]
```

p(95) < 250 ms 235.79 ms
p(99) < 1000 ms 432.52 ms
failure rate < 1% 0.00%

Baseline Load Test: The system was tested with up to 25 concurrent virtual users over approximately 3.5 minutes. The test generated 14 340 HTTP requests at an average throughput of 62.7 requests/second.

No HTTP request failures occurred. The 95th-percentile response time was 235.79 ms and the 99th-percentile response time was 432.52 ms. During observation, container resource utilisation remained low, with the backend using approximately 295 MiB of memory and 0.20% CPU at the sampled point.

50 VUs

```
stages: [  
  { duration: '30s', target: 10},  
  { duration: '1m', target: 10},  
  { duration: '30s', target: 50},  
  { duration: '1m', target: 50},  
  { duration: '30s', target: 0},  
]
```

Metric	25 VUs	50 VUs	Interpretation
Throughput	62.7 req/s	107.2 req/s	Increased substantially
Average	237.72 ms	238.16 ms	Essentially unchanged
p95	235.79 ms	229.76 ms	Essentially unchanged
p99	432.52 ms	466.26 ms	Small increase

Failure rate	0%	0%	Excellent	
Max	2.94 s	2.95 s	Essentially unchanged	

p95 < 250 ms 229.76 ms
p99 < 1000 ms 466.26 ms
failure < 1% 0%

The system successfully supported 50 concurrent virtual users, processing 24 548 requests at an average throughput of 107.2 requests/second. No HTTP request failures were observed. Response-time performance remained stable, with an average latency of 238.16 ms, p95 latency of 229.76 ms, and p99 latency of 466.26 ms. The results indicate that the single-instance deployment comfortably handled the tested 50-VU workload without significant performance degradation.

100 VUs

Metric	50 VUs	100 VUs	Interpretation	
-----	-----:	-----:	-----	
Requests	24,548	37,240	Higher	
Throughput	107.2 req/s	142.6 req/s	Higher	
Average	238.16 ms	243.13 ms	Slight increase	
Median	217.33 ms	217.25 ms	Essentially unchanged	
p95	229.76 ms	237.80 ms	Slight increase	
p99	466.26 ms	640.63 ms	Noticeable increase	
Failure rate	0%	0%	Same	
Maximum	2.95 s	19.34 s	Found outlier	

p95 < 250 ms
p99 < 1000 ms
failure < 1%

25 → 50 → 100 VUs did not cause significant degradation in p95 latency.

Testing using GET /api/timers/active

Full Endpoint: GET <https://momentlyapp.org/api/timers/active>